Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A process has ΔU=-15 J and W_{by}=21 J. Find Q and decide whether heat enters or leaves.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$

$Q_{absorbed}$ = fraction absorbed × energy available.

Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?



A B

Marzano check - circle: ① ② ③

04 Apply the physics

An atmospheric-entry simulation starts a hypothetical 10 kg rock at 15 km/s. A supplied heating-and-vaporization requirement is 8.6×10^6 J/kg. Calculate its kinetic energy, total requirement and their ratio. Label possible destinations for the energy; explain why the ratio alone cannot predict survival.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

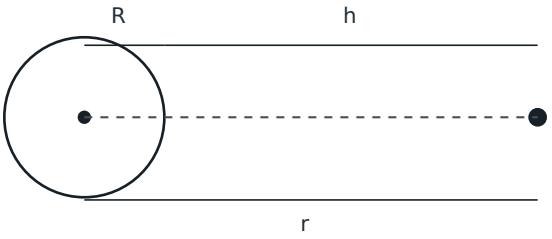
05 Explain your reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

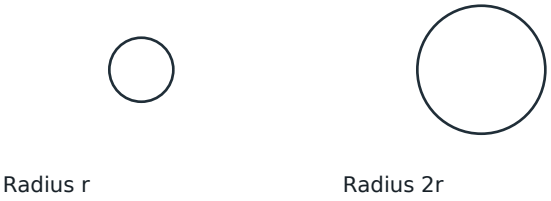
For fixed GM, launch radius becomes 16 times as large. Compare ideal escape speeds and name the equation. Label the distances or energy levels used in your reasoning.



Marzano check - circle: 1 2 3

07 Exit check

If radius doubles at fixed density, derive the area-to-mass multiplier and explain why kinetic-energy-per-mass alone cannot predict a size threshold for survival. Label energy entering, leaving or stored on the system model.



Radius r Radius 2r

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg sample has 20 kJ deposited as internal energy. With effective c=500 J/(kg·K), estimate ΔT if no phase change or loss occurs. State why this is not a survival threshold. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3